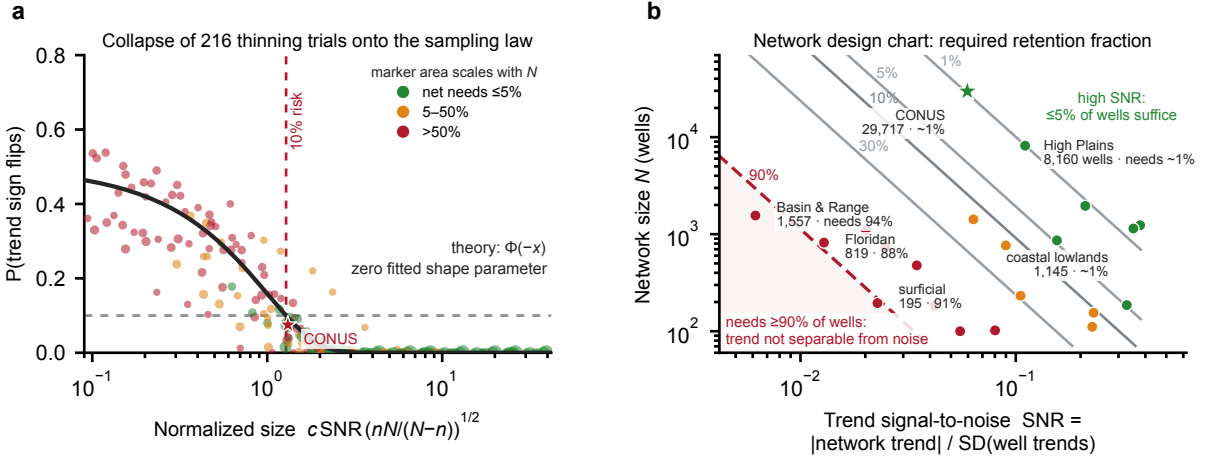

a
Collapse of 216 thinning trials onto the sampling law
marker area scales with N
net needs ≤5%
5–50%
>50%
10% risk
theory: Φ(−x)
zero fitted shape parameter
CONUS
P(trend sign flips)
Normalized size c SNR(nN/(N−n))^1/2
b
Network design chart: required retention fraction
1%
5%
10%
30%
CONUS
29,717 · ~1%
high SNR:
≤5% of wells suffice
High Plains
8,160 wells · needs ~1%
90%
Basin & Range
1,557 · needs 94%
Floridan
819 · 88%
coastal lowlands
1,145 · ~1%
surficial
195 · 91%
needs ≥90% of wells:
trend not separable from noise
Network size N (wells)
Trend signal-to-noise SNR =
|network trend| / SD(well trends)